

Today's Content:-

- find 2 unique elements
- find 2 missing elements
- Calculate sum of XOR of all pairs
- find max $A \oplus B$

Q1 Given N elements, every element repeats twice, except 2 unique elements. Find the 2 unique elements.

e.g. $arr[6] = \{3, 6, 4, 4, 3, 8\} \rightarrow 6, 8$

$arr[4] = \{4, 9, 9, 8\} \rightarrow 4, 8$

Idea1:- Hash Map.

T.C = $O(N)$

S.C = $O(N)$

Idea2:- Sort the array.

T.C = $O(N \log N + N)$

S.C = $O(1)$

Idea3:- XOR of all elements.

$arr[6] = \{3, 6, 4, 4, 3, 8\} \rightarrow 6 \wedge 8 = 14$

$\nearrow a$
 $\searrow b$

XOR of all elements = XOR of 2 unique elements

e.g. $arr[] :-$

10	8	8	9	12	9	6	11	10	6	12	17
(1010)	(1000)	(1000)	(1100)	(1100)	(1001)	(0110)	(1011)	(1010)	(0110)	(1100)	(1001)

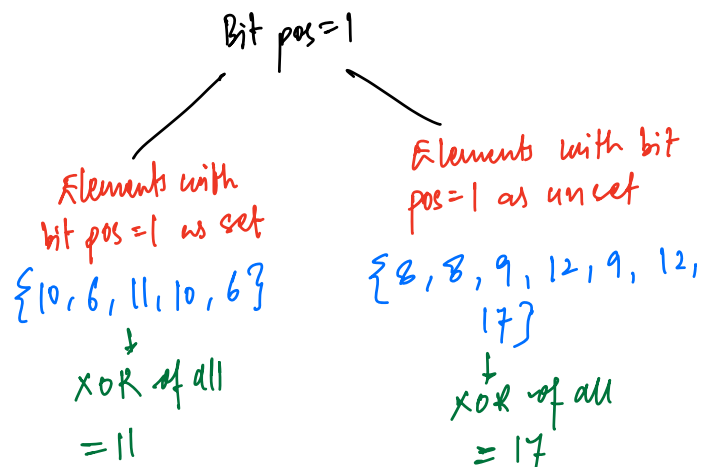
$11 \wedge 17$

$\begin{array}{r} 4 \ 3 \ 2 \ 1 \ 0 \\ 11: \ 0 \ 1 \ 0 \ 1 \ 1 \end{array}$

$\begin{array}{r} 17: \ 1 \ 0 \ 0 \ 0 \ 1 \end{array}$

$11 \wedge 17 = 11010 = 26$

At bit pos:- 1, 3, 4



Pseudo code:-

Check Wednesday's Session.

Q2 Given N array elements, array contains all elements from $[1, N+2]$ except 2 elements.

Find 2 missing elements.

$\rightarrow N=4 \rightarrow [1-6]$
 $arr[4] = \{3, 6, 1, 4\} \rightarrow \{2, 5\}$ missing elements

$\rightarrow N=5 \rightarrow [1-7]$
 $arr[5] = \{1, 6, 4, 7, 5\} \rightarrow \{2, 3\}$

ideas:-

1) Use hashset and find the missing elements.

$T.C = O(N)$; $S.C = O(N)$

2) Sort the array and get the missing elements.

$T.C = O(N \log N + N)$
 $S.C = O(1)$

3) Use a boolean array of size $N+2$.

$T.C = O(N)$; $S.C = O(N)$

$\hookrightarrow$ slightly better than Approach 1.

$\rightarrow N=5 \{1, 2, 3, 4, 5, 6, 7\}$ $\rightarrow$ Expected $T.C = O(N)$
 $arr[5] = \{1, 6, 4, 7, 5\}$ $S.C = O(1)$

$\downarrow$
Every element repeats twice
except 2 unique elements.

Q3 Given N elements. Calculate sum of XOR of all pairs.

$N \geq 0$
 $arr[i] \geq 0$

$arr[5] = \{3, 5, 6, 8, 2\}$

Ideas:-

1) Brute force:- Sum of all XOR pairs

2)

$3 \wedge 3$	$3 \wedge 5$	$3 \wedge 6$	$3 \wedge 8$	$3 \wedge 2$
$5 \wedge 3$	$5 \wedge 5$	$5 \wedge 6$	$5 \wedge 8$	$5 \wedge 2$
$6 \wedge 3$	$6 \wedge 5$	$6 \wedge 6$	$6 \wedge 8$	$6 \wedge 2$
$8 \wedge 3$	$8 \wedge 5$	$8 \wedge 6$	$8 \wedge 8$	$8 \wedge 2$
$2 \wedge 3$	$2 \wedge 5$	$2 \wedge 6$	$2 \wedge 8$	$2 \wedge 2$

3) Idea:- Write Binary representation of all elements.

```

sum = 0
TC = O(N^2); SC = O(1)
for (i = 0; i < N; i++) {
    for (j = 0; j < N; j++) {
        sum = sum + arr[i] ^ arr[j];
    }
}

```

```

sum = 0
for (i = 0; i < N; i++) {
    for (j = i + 1; j < N; j++) {
        sum = sum + arr[i] ^ arr[j];
    }
}
ans = sum * 2;

```

TC = O(N^2); SC = O(1)

2: 0 0 1 0

3: 0 0 1 1

5: 0 1 0 1

6: 0 1 1 0

8: 1 0 0 0

Idea 3:-

	C_3	C_2	C_1	C_0
	2^3	2^2	2^1	2^0
$3^1 5$:	0	1	1	0
$3^1 6$:	0	1	0	1
$3^1 8$:	1	0	1	1
$3^1 2$:	0	0	0	1
$5^1 6$:	0	0	1	1
$5^1 8$:	1	1	0	1
$5^1 2$:	0	1	1	1
$6^1 8$:	1	1	1	0
$6^1 2$:	0	1	0	0
$8^1 2$:	1	0	1	0
	4	6	6	6
	2^3	2^2	2^1	2^0

idea:- At i th bit position $\rightarrow C_i$ number of pairs have i th bit pos. as set

$C_0 = 6$
 $C_1 = 6$
 $C_2 = 6$
 $C_3 = 4$

$$\text{sum} = \sum_{i=0}^3 C_i * 2^i$$

$$\text{ans} = 2 * \text{sum}$$

T.C. is worse:-

$$\begin{aligned} \text{T.C.} &= 31 * N^2 \\ &= O(N^2) \\ \text{S.C.} &= O(1) \end{aligned}$$

$$2^3 * 4 + 2^2 * 6 + 2^1 * 6 + 2^0 * 6$$

$$32 + 24 + 12 + 6 = 74$$

$$\text{total} = 2 * 74 = 148$$

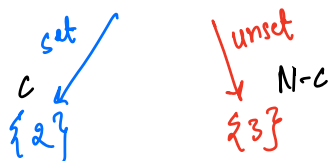
2: 0 0 1 0 3: 0 0 1 1 5: 0 1 0 1 6: 0 1 1 0 8: 1 0 0 0

Idea 4:-

idea 4:- XOR $\begin{cases} \rightarrow 0 & \text{same} \\ \rightarrow 1 & \text{different} \end{cases}$

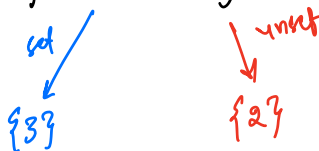
	2^3	2^2	2^1	2^0
$3 \wedge 5$:	0	1	1	0
$3 \wedge 6$:	0	1	0	1
$3 \wedge 8$:	1	0	1	1
$3 \wedge 2$:	0	0	0	1
$5 \wedge 6$:	0	0	1	1
$5 \wedge 8$:	1	1	0	1
$5 \wedge 2$:	0	1	1	1
$6 \wedge 8$:	1	1	1	0
$6 \wedge 2$:	0	1	0	0
$8 \wedge 2$:	1	0	1	0
	4	6	6	6
	2^3	2^2	2^1	2^0
	$6 \wedge 2^3, 6 \wedge 2^2$			

In given array, how many elements have 0th bit set?



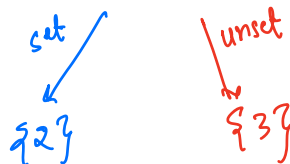
No. of pairs = $2 \times 3 = 6$ pairs.

In given array, how many elements have 1th bit set?



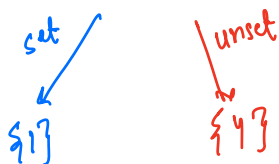
6 pairs.

In given array, how many elements have 2th bit set?



6 pairs.

In given array, how many elements have 3th bit set?



4 pairs.

Pseudo code:-

sum = 0;

for (i = 0; i <= 30; i++) {

 c = 0;

 for (j = 0; j < N; j++) {

 if (checkBit(arr[j], i)) {

 c++;

 }

 sum = sum + c * (N - c) * (1 < i)

}

return 2 * sum;

$$T.C = 31 * N \\ = O(N)$$

$$S.C = O(1)$$

Q. Given N array elements, choose 2 elements (i, j) such that $i \neq j$ and $A[i] \& A[j]$ is max.
 ↳ Bitwise & → **GOOGLE QUESTION**

e.g. arr[3] = { 27, 18, 20 }

(27, 18)
 18: 1 0 0 1 0
 27: 1 1 0 1 0

1 0 0 1 0 = 14

Ans = 14

(27, 20)
 27: 1 1 0 1 1
 20: 1 0 1 0 0

1 0 0 0 0 = 16

(18, 20)
 18: 1 0 0 1 0
 20: 1 0 1 0 0

1 0 0 0 0 = 16

e.g. arr[4] = { 21, 18, 24, 17 }

21: 1 0 1 0 1
 18: 1 0 0 1 0
 24: 1 1 0 0 0
 17: 1 0 0 0 1

21: 1 0 1 0 1
 17: 1 0 0 0 1
1 0 0 0 1 = 17

21: 1 0 1 0 1
 24: 1 1 0 0 0
1 0 0 0 0 = 16 X

1) List → { 17, 18, 21, 24 } → X

2) Max value in the array should be 1 of the elements that we do & of to get the max $A[i] \& A[j]$ → X

5) Check all pairs and get max 'L'.

T.C = $O(N^2)$

S.C = $O(1)$

e.g. arr[7] = {26, ~~13~~, ~~23~~, 28, 27, ~~7~~, 25}

26:-

	4	3	2	1	0
	1	1	0	1	0

13:-

0	1	1	0	1
---	---	---	---	---

 → 0

23:-

1	0	1	1	1
---	---	---	---	---

 → 0

28:-

1	1	1	0	0
---	---	---	---	---

27:-

1	1	0	1	1
---	---	---	---	---

7:-

0	0	1	1	1
---	---	---	---	---

 → 0

25:-

1	1	0	0	1
---	---	---	---	---

1	1	0	1	0
---	---	---	---	---

→ maximum AND
(26 & 27)

25	22	21	20
		↓	
		23-1	

Pseudo Code:-

```
int ans = 0;
```

```
for (i = 30; i >= 0; i--) {
```

```
    c = 0
```

```
    for (j = 0; j < N; j++) {
```

```
        if (checkBit(arr[j], i) {
```

```
            c++;
```

```
        }
```

```
    if (c >= 2) {
```

```
        ans = ans + (1 << i);
```

```
        for (j = 0; j < N; j++) {
```

```
            if (!checkBit(arr[j], i) {
```

```
                arr[j] = 0;
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

```
return ans;
```

$$T.C = \sum_{i=0}^{30} [N+1]$$

$$= 62N = O(N)$$

$$S.C = O(1)$$